

DATANEX INTERNSHIP PROGRAM

WEEK 2 TASK REPORT

Linux Command Line – OverTheWire Bandit

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Task	Completed Bandit Levels 0–5

Submitted as part of the DATANEX Cybersecurity Internship

1. Task Overview

The Week 2 task focused on Linux command-line fundamentals through OverTheWire: Bandit, a legal and controlled cybersecurity practice environment. The objective was to complete Levels 0–5 and demonstrate practical use of SSH, file navigation, file reading, hidden-file discovery, file-type identification, and recursive file searching.

2. Objectives

- Connect to a remote Linux system securely using SSH.
- Navigate directories and inspect files from the terminal.
- Read files with cat and handle filenames containing special characters and spaces.
- Identify hidden files using ls -la.
- Identify readable files using the file command.
- Search recursively for files using specific properties with find.
- Understand how command-line skills support cybersecurity and remote-system administration.

3. Tools and Environment

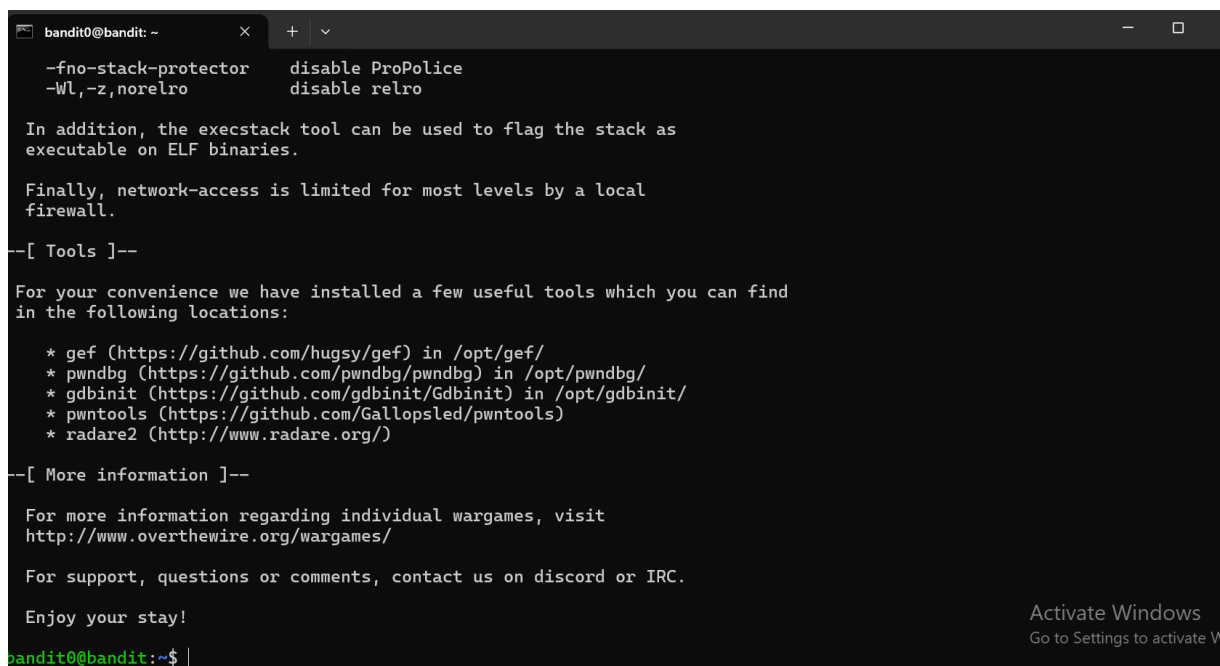
- Windows PowerShell / Terminal
- SSH (Secure Shell)
- OverTheWire Bandit server
- Linux command-line utilities: ssh, ls, cd, cat, file, find

4. Implementation and Evidence

Level 0 – Initial SSH Connection

Connected to the Bandit server using SSH. The command used was `ssh bandit0@bandit.labs.overthewire.org -p 2220`. The default Level 0 credentials were used to establish the session.

Result: The successful prompt `bandit0@bandit:~$` confirmed that the remote Linux shell was active.



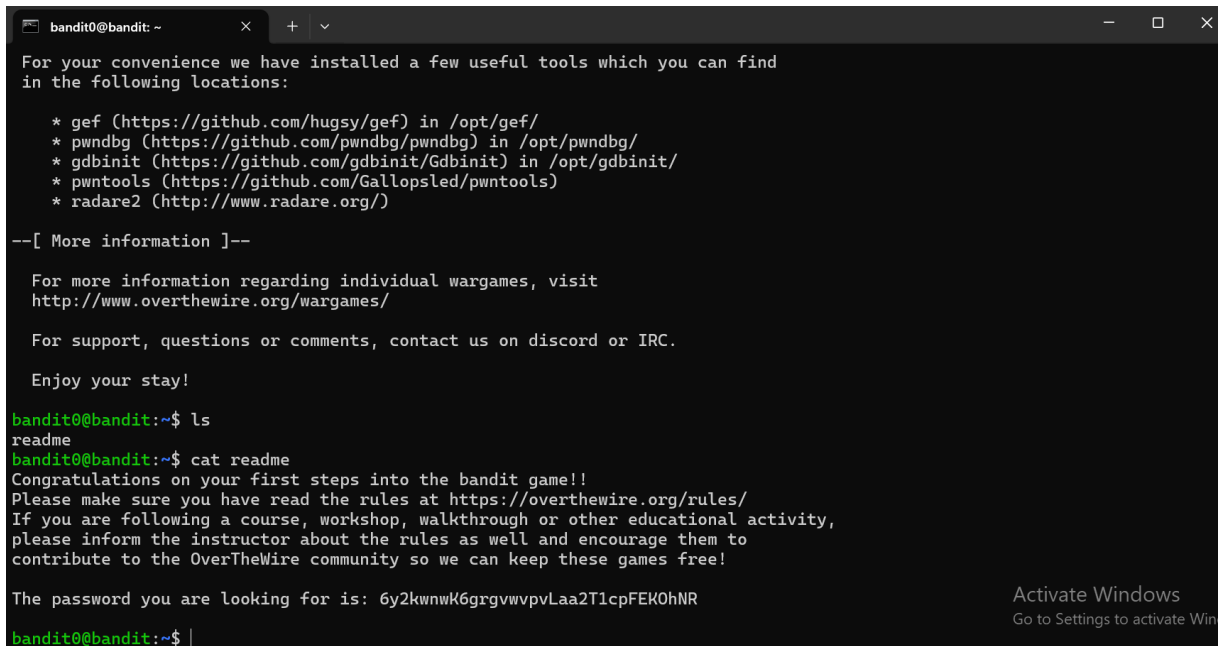
```
bandit0@bandit: ~  
-fno-stack-protector  disable ProPolice  
-Wl,-z,norelro        disable relro  
  
In addition, the execstack tool can be used to flag the stack as  
executable on ELF binaries.  
  
Finally, network-access is limited for most levels by a local  
firewall.  
  
--[ Tools ]--  
  
For your convenience we have installed a few useful tools which you can find  
in the following locations:  
  
* gef (https://github.com/hugsy/gef) in /opt/gef/  
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/  
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/  
* pwntools (https://github.com/Gallopsled/pwntools)  
* radare2 (http://www.radare.org/)  
  
--[ More information ]--  
  
For more information regarding individual wargames, visit  
http://www.overthewire.org/wargames/  
  
For support, questions or comments, contact us on discord or IRC.  
  
Enjoy your stay!  
bandit0@bandit:~$
```

Evidence: WK2_Screenshot_01_Bandit_Level0_Login.png

Level 0 → Level 1 – Reading the README

Used `ls` to list the home directory and `cat readme` to display the README contents. The password for the next level was obtained from the file.

Result: This demonstrated basic directory listing and text-file reading.



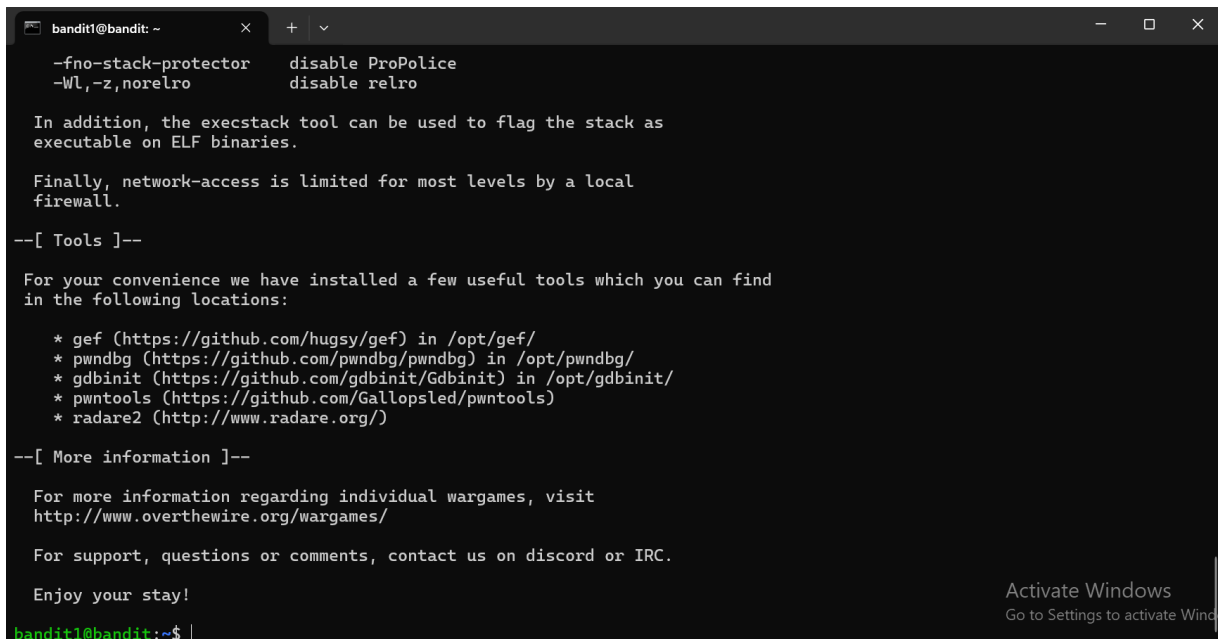
```
bandit0@bandit: ~  
For your convenience we have installed a few useful tools which you can find  
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--[ More information ]--  
  
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http://www.overthewire.org/wargames/  
  
For support, questions or comments, contact us on discord or IRC.  
  
Enjoy your stay!  
bandit0@bandit:~$ ls  
readme  
bandit0@bandit:~$ cat readme  
Congratulations on your first steps into the bandit game!!  
Please make sure you have read the rules at https://overthewire.org/rules/  
If you are following a course, workshop, walkthrough or other educational activity,  
please inform the instructor about the rules as well and encourage them to  
contribute to the OverTheWire community so we can keep these games free!  
  
The password you are looking for is: 6y2kwnwK6grgvwvpvLaa2T1cpFEK0hNR  
bandit0@bandit:~$ |
```

Evidence: WK2_Screenshot_02_Level0_Readme.png

Level 1 – Login and Special Filename

Connected as `bandit1` using SSH. The target file was named `-`, which can be interpreted specially by command-line programs. The file was safely read with `cat ./-`.

Result: Using `./-` explicitly identified the file in the current directory.



```
bandit1@bandit: ~  
-fno-stack-protector  disable ProPolice  
-Wl,-z,norelro         disable relro  
  
In addition, the execstack tool can be used to flag the stack as  
executable on ELF binaries.  
  
Finally, network-access is limited for most levels by a local  
firewall.  
  
--[ Tools ]--  
  
For your convenience we have installed a few useful tools which you can find  
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http://www.overthewire.org/wargames/  
  
For support, questions or comments, contact us on discord or IRC.  
  
Enjoy your stay!  
bandit1@bandit:~$ |
```

Evidence: WK2_Screenshot_03_Bandit_Level1_Login.png

```
bandit1@bandit: ~  
executable on ELF binaries.  
  
Finally, network-access is limited for most levels by a local  
firewall.  
  
--[ Tools ]--  
  
For your convenience we have installed a few useful tools which you can find  
in the following locations:  
  
* gef (https://github.com/hugsy/gef) in /opt/gef/  
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/  
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/  
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For more information regarding individual wargames, visit  
http://www.overthewire.org/wargames/  
  
For support, questions or comments, contact us on discord or IRC.  
  
Enjoy your stay!  
bandit1@bandit:~$ ls  
-  
bandit1@bandit:~$ cat ./-  
PK8fYLZg2hnHSz83p1BL1iEPKdD3QT0B  
bandit1@bandit:~$ |
```

Evidence: WK2_Screenshot_04_Level1_Special_Filename.png

Level 2 – Filename with Spaces

Connected as bandit2 and inspected the directory. The target filename was --spaces in this filename--. The exact filename was read using cat "./--spaces in this filename--".

Result: Quoting the filename prevented spaces and leading hyphens from being interpreted incorrectly.

```
bandit2@bandit: ~  
-fno-stack-protector  disable ProPolice  
-Wl,-z,norelro        disable relro  
  
In addition, the execstack tool can be used to flag the stack as  
executable on ELF binaries.  
  
Finally, network-access is limited for most levels by a local  
firewall.  
  
--[ Tools ]--  
  
For your convenience we have installed a few useful tools which you can find  
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--[ More information ]--  
  
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For support, questions or comments, contact us on discord or IRC.  
  
Enjoy your stay!  
bandit2@bandit:~$ |
```

Evidence: WK2_Screenshot_05_Bandit_Level2_Login.png

```
bandit2@bandit: ~  
Finally, network-access is limited for most levels by a local  
firewall.  
  
--[ Tools ]--  
  
For your convenience we have installed a few useful tools which you can find  
in the following locations:  
  
* gef (https://github.com/hugsy/gef) in /opt/gef/  
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/  
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/  
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--[ More information ]--  
  
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For support, questions or comments, contact us on discord or IRC.  
  
Enjoy your stay!  
  
bandit2@bandit:~$ ls  
--spaces in this filename--  
bandit2@bandit:~$ cat "spaces in this filename"  
cat: 'spaces in this filename': No such file or directory  
bandit2@bandit:~$ cat "./--spaces in this filename--"  
7ZZ2LFrykP2zEyvBl4m3clcl7tGYJPME  
bandit2@bandit:~$ |
```

Evidence: WK2_Screenshot_06_Level2_Filename_With_Spaces.png

Level 3 – Hidden File Discovery

Connected as bandit3, entered the inhere directory, and used `ls -la` to reveal hidden entries. The hidden file was named `...Hiding-From-You` and was read with `cat ./...Hiding-From-You`.

Result: The task demonstrated that hidden files are not shown by a normal ``ls`` listing.

```
bandit3@bandit: ~  
-fno-stack-protector    disable ProPolice  
-Wl,-z,norelro          disable relro  
  
In addition, the execstack tool can be used to flag the stack as  
executable on ELF binaries.  
  
Finally, network-access is limited for most levels by a local  
firewall.  
  
--[ Tools ]--  
  
For your convenience we have installed a few useful tools which you can find  
in the following locations:  
  
* gef (https://github.com/hugsy/gef) in /opt/gef/  
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--[ More information ]--  
  
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http://www.overthewire.org/wargames/  
  
For support, questions or comments, contact us on discord or IRC.  
  
Enjoy your stay!  
  
bandit3@bandit:~$ |
```

Evidence: WK2_Screenshot_07_Bandit_Level3_Login.png

```
bandit3@bandit: ~/inhere
in the following locations:

* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)

--[ More information ]--

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For support, questions or comments, contact us on discord or IRC.

Enjoy your stay!

bandit3@bandit:~$ ls
inhere
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls -la
total 12
drwxr-xr-x 2 root root 4096 Jun 24 14:59 .
drwxr-xr-x 3 root root 4096 Jun 24 14:59 ..
-rw-r----- 1 bandit4 bandit3 33 Jun 24 14:59 ...Hiding-From-You
bandit3@bandit:~/inhere$ cat .hidden
cat: .hidden: No such file or directory
bandit3@bandit:~/inhere$ cat ...Hiding-From-You
xzTXqlrDJQVVAzdv5cHq1TQytTWufAMq
bandit3@bandit:~/inhere$ |
```

Evidence: WK2_Screenshot_08_Level3_Hidden_File.png

Level 4 – Identifying Human-Readable Data

Connected as bandit4, entered inhere, and examined the files with file ./* . The output identified ./file07 as ASCII text. It was then read with cat ./file07.

Result: The `file` utility helped distinguish readable text from binary and other data.

```
bandit4@bandit: ~
-fno-stack-protector  disable ProPolice
-Wl,-z,norelro        disable relro

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

Finally, network-access is limited for most levels by a local
firewall.

--[ Tools ]--

For your convenience we have installed a few useful tools which you can find
in the following locations:

* gef (https://github.com/hugsy/gef) in /opt/gef/
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Enjoy your stay!

bandit4@bandit:~$ |
```

Evidence: WK2_Screenshot_09_Bandit_Level4_Login.png

```
bandit4@bandit: ~/inhere
bandit4@bandit:~$ ls
inhere
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls -la
total 48
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file00
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file01
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file02
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file03
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file04
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file05
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file06
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file07
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file08
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file09
drwxr-xr-x 2 root root 4096 Jun 24 14:59 .
drwxr-xr-x 3 root root 4096 Jun 24 14:59 ..
bandit4@bandit:~/inhere$ file ./-file00: data
./-file01: data
./-file02: OpenPGP Secret Key
./-file03: data
./-file04: data
./-file05: data
./-file06: Non-ISO extended-ASCII text, with NEL line terminators
./-file07: ASCII text
./-file08: data
./-file09: data
bandit4@bandit:~/inhere$
```

Evidence: WK2_Screenshot_10_Level4_File_Type_Check.png

```
bandit4@bandit: ~/inhere
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls -la
total 48
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file00
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file01
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file02
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file03
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file04
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file05
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file06
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file07
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file08
-rw-r----- 1 bandit5 bandit4 33 Jun 24 14:59 -file09
drwxr-xr-x 2 root root 4096 Jun 24 14:59 .
drwxr-xr-x 3 root root 4096 Jun 24 14:59 ..
bandit4@bandit:~/inhere$ file ./-file00: data
./-file01: data
./-file02: OpenPGP Secret Key
./-file03: data
./-file04: data
./-file05: data
./-file06: Non-ISO extended-ASCII text, with NEL line terminators
./-file07: ASCII text
./-file08: data
./-file09: data
bandit4@bandit:~/inhere$ cat ./-file07
6C7h9GD8M6ai5nr7wo1RonrzFjj9yIrG
bandit4@bandit:~/inhere$
```

Evidence: WK2_Screenshot_11_Level4_Password.png

Level 5 – Recursive Search by File Properties

Connected as bandit5 and entered inhere. The directory contained many subdirectories. The command `find . -type f -size 1033c ! -executable` searched recursively for a regular file that was exactly 1033 bytes and not executable. The result was `./maybehere07/.file2`, which was read using `cat ./maybehere07/.file2`.

Result: This demonstrated practical recursive searching using file type, size, and executable-status conditions.

```
bandit5@bandit: ~  
-fno-stack-protector    disable ProPolice  
-Wl,-z,norelro          disable relro  
  
In addition, the execstack tool can be used to flag the stack as  
executable on ELF binaries.  
  
Finally, network-access is limited for most levels by a local  
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--[ Tools ]--  
  
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* radare2 (http://www.radare.org/)  
  
--[ More information ]--  
  
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For support, questions or comments, contact us on discord or IRC.  
  
Enjoy your stay!  
bandit5@bandit:~$ |
```

Evidence: WK2_Screenshot_12_Bandit_Level5_Login.png

```
bandit5@bandit: ~/inhere  
bandit5@bandit:~$ ls  
inhere  
bandit5@bandit:~$ cd inhere  
bandit5@bandit:~/inhere$ ls -la  
total 88  
drwxr-x--- 22 root bandit5 4096 Jun 24 14:59 .  
drwxr-xr-x  3 root root    4096 Jun 24 14:59 ..  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere00  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere01  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere02  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere03  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere04  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere05  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere06  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere07  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere08  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere09  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere10  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere11  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere12  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere13  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere14  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere15  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere16  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere17  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere18  
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere19  
bandit5@bandit:~/inhere$ find . -type f -size 1033c ! -executable  
./maybehere07/.file2  
bandit5@bandit:~/inhere$ |
```

Evidence: WK2_Screenshot_13_Level5_Find_Command.png


```
bandit5@bandit: ~/inhere
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere03
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere04
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere05
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere06
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere07
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere08
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere09
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere10
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere11
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere12
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere13
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere14
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere15
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere16
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere17
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere18
drwxr-x--- 2 root bandit5 4096 Jun 24 14:59 maybehere19
bandit5@bandit:~/inhere$ find . -type f -size 1033c ! -executable
./maybehere07/.file2
bandit5@bandit:~/inhere$ cat ./maybehere07/.file2
pXa26xhMWaC2SvDotA4r9EgZku0eSBW
bandit5@bandit:~/inhere$
```

Evidence: WK2_Screenshot_14_Level5_Password.png

5. Linux Commands Learned

Command	Purpose
ssh	Connect to a remote system using the SSH protocol.
ls	List files and directories.
ls -la	List files including hidden entries with detailed information.
cd	Change the current working directory.
cat	Display the contents of a file.
file	Identify the type/content classification of a file.
find	Search recursively for files using conditions such as type, size, and permissions.
./	Explicitly refer to a path in the current directory, useful for filenames beginning with `.`.

6. Cybersecurity Relevance

Command-line proficiency is important in cybersecurity because security analysts, administrators, and penetration testers frequently work with remote systems, logs, files, and security tools. SSH provides remote command-line access, while commands such as find, file, ls, and cat support basic system inspection and evidence discovery.

7. Challenges and Solutions

- Special filename -: solved by explicitly using ./-.
- Filename containing spaces and leading hyphens: solved by quoting the exact filename and using an explicit relative path.
- Hidden-file discovery: solved with ls -la.
- Identifying readable data among multiple files: solved with file ./*.
- Searching many directories: solved with find and multiple conditions.

8. Skills Gained

- Basic Linux terminal navigation
- Remote Linux access using SSH
- Command-line file handling
- Hidden-file discovery
- File-type analysis
- Recursive file searching
- Practical troubleshooting of command syntax

9. Security and Credential Handling Note

The screenshots contain credentials used inside the OverTheWire Bandit training environment. They are included as practical evidence for the internship report. These credentials should not be published in public LinkedIn posts or public repositories. Public-facing screenshots should be cropped or redacted before posting.

10. Conclusion

The Week 2 task was completed by successfully working through OverTheWire Bandit Levels 0–5. The practical exercises established a foundation in Linux command-line usage, SSH access, file handling, hidden-file discovery, file identification, and recursive searching. These are foundational skills for further cybersecurity activities involving Linux systems and remote environments.